Facet

Abstract: A platform that acts as an abstraction layer between application backends and end users. Developers expose their application's API through the platform, while users interact with an AI agent that analyzes the available resources and generates a personalized frontend tailored to their workflow, without requiring changes to the backend.

Code Idea:

Developers integrate their applications with Facet by registering their API through the Facet platform. Each application includes a high-level description explaining its purpose, while every endpoint is accompanied by a developer-defined description explaining what it does and how it should be used. This metadata is maintained independently from the backend implementation, allowing developers to improve the AI's understanding of their application without modifying their source code.

Applications are published on the Facet platform, where users can browse available applications and choose one to use. After selecting an application, the user describes how they would like to interact with it—for example, requesting a Kanban board, a calendar view, or a minimal task-focused interface.

The Facet agent reads the application's metadata, including its abstract and endpoint descriptions, understands the application's capabilities, and generates a frontend that satisfies the user's request while exposing the complete functionality of the underlying application. The generated frontend communicates with the original backend exclusively through the registered API endpoints.

Example JSON of an application:

```
{
  "name": "TaskFlow",
  "abstract": "A task management application for teams.",

  "endpoints": [
    {
      "method": "GET",
      "path": "/tasks",
      "description": "Returns all tasks assigned to the current user."
    },
    {
      "method": "POST",
      "path": "/tasks",
      "description": "Creates a new task."
    }
```

  ]
}

Prototype:

For the prototype, we will not implement the complete developer library or backend integration. Instead, we will build the Facet platform itself.

The developer dashboard will allow developers to create applications, write an application abstract, and register API endpoints together with their descriptions. These endpoint definitions will be stored as structured metadata that the AI agent can consume.

The user interface will allow users to browse the available applications, select one, and describe the type of interface they would like to use. The AI agent will read the application's metadata and generate a frontend that follows the user's request while ensuring that every feature exposed by the registered API remains accessible.

Prototype Website

The website will be divided into 3 main pages: Landing page, Developer dashboard and application marketplace.

Tech Stack: Typescript NodeJS + React
Style: Minimalistic modern, darker colors and gradients
Database: Fake it for now, just make everything stored and loaded from a local JSON file.
Everything will be designed only to run on localhost for now.

Developer Dashboard

It should contain the following left-sided tabs: Subscription, Applications, API key page.

Subscription: Simple subscription page where u can choose between a list of subscriptions each with different price and different rate limits. Essentially the developer pays for a subscription that will have a limit of monthly API calls, this is our main income source (e.g. Personal, Pro, Max, Enterprise tier)

API key page: Just lets u generate private keys that u will use when coding the back-end to connect to your applications, it allows u to creat a new one, define to what apps this key gives u access, if it expires etc and allows u to delete/re-genereate one and whatnot. Just make it all fake and store the things locally.
Applications: The main one that should be used by default. It shows a list of the applications u own and a button to add a new one. U can also press an edit button on each application. The following configurations for an application contain: The name, an abstract describing the application, a list of endpoints that u can create, change and delete. Each endpoint should allow u to select: type of request, the tuple that it receives/returns, a description of what it has.

Application MarketPlace

A simple list with all the applications that exist and a browsing bar so u can search for name. For the prototype make it simple so that u can just search by the name locally (using the json file) and that's it. Make it so that if u click an application it moves to the website of this specific applications: it should contain the name, abstract and a prompt field where u can describe your app and then press a button that says "Build". This build button will generate another end-point on the website which contains ur AI generate web application that connects to this back-end. I want u to make it so that the web page will use an HTML page generate by an AI agent (Set it up to use chatgpt and just leave the .env key empty which I will write down myself).

Landing Page

Modern catchy, marketing-like page promoting the application. With a button to go directly to the application marketplace.